Notes: Category Theory in Context

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These notes are intended as a concise overview of the book Category Theory in Context by Emiliy Riehl.

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1 Categories, functors, natural transformations

1.1 Abstract and concrete categories

Definition 1.1.1 (Category)**.** Notation: $\mathcal{C}, \mathcal{D}, \mathcal{E}, \ldots$

Remark 1.1.2 (Correspondence between objects and identity morphisms)**.**

Example 1.1.3 (Concrete categories)**.** $\mathcal{S}et$-based categories:

- $\mathcal{S}et$, $\mathcal{T}op$;
- $\mathcal{S}et_*$, $\mathcal{T}op_*$ (pointed);
- $\mathcal{G}rp$, $\mathcal{R}ing$, $\mathcal{F}ield$;
- $\mathcal{A}b\mathcal{G}rp$, $\mathcal{V}ect_{\mathbf{k}}$, $\mathcal{M}od_R$;
- $\mathcal{G}raph$, $\mathcal{D}ir\mathcal{G}raph$;
- $\mathcal{M}an$, $\mathcal{M}eas$ – manifolds and measurable spaces;

– $\mathcal{P}oset$;

– $\mathcal{C}h_R$ – chain complexes;

– $\mathcal{M}odel_{\mathbb{T}}$ – σ-structures that model $\mathbb{T}$ (see universal algebra).

Remark. The category $\mathcal{S}et$ implicitly depends on the underlying set theory.

Example 1.1.4 (Abstract categories)**.**

– $\mathcal{M}at_R$ – objects are integers, arrows $m \to n$ are $m \times n$ matrices with values in R;

– $\mathcal{B}G$;

– $(P, \leq)$ and ordinal $\alpha = \{\beta \mid \beta < \alpha\}$ as categories;

– $\mathcal{H}tpy$, $\mathcal{H}tpy_*$, $\mathcal{M}easure$.

Remark 1.1.5 (Russel's paradox)**.**

Definition 1.1.6 (Small category)**.** A category with only a set's worth of arrows.

Definition 1.1.7 (Locally small category)**.** A category with only set's worth of arrows between any pair of objects.

Definition 1.1.8 (Isomorphism, automorphism, endomorphism)**.**

Example 1.1.9 (Isomorphisms in categories)**.**

Definition 1.1.10 (Groupoid)**.** A groupoid is a category in which every morphism is an isomorphism.

Example 1.1.11 (Examples of groupoids)**.**

Definition (Subcategory)**.**

Lemma 1.1.12 (Maximal groupoid)**.** Any category $\mathcal{C}$ contains a maximal groupoid, the subcategory generated by isomorphism of $\mathcal{C}$.

Definition (Slice categories $c/\mathcal{C}$ and $\mathcal{C}/c$)**.**

1.2 Duality

Definition 1.2.1 (Opposite category $\mathcal{C}^{\mathrm{op}}$)**.**

Remark. This duality is purely syntactic. Its usefulness becomes apparent when a) considering opposite constructions, b) working with functors.

Example 1.2.2 (Opposite categories)**.** $\mathcal{M}at_R^{\mathrm{op}}$, $(P, \leq)^{\mathrm{op}}$, $(\mathcal{B}G)^{\mathrm{op}}$.

Lemma 1.2.3. The following are equivalent (in a locally small $\mathcal{C}$):

1. $f : x \to y$ is an iso in $\mathcal{C}$.
2. For all $c : \mathrm{obj}(\mathcal{C})$, post-composition with f defines a bijection

$$f_* : \mathcal{C}(c, x) \to \mathcal{C}(c, y).$$

3. For all $c : \mathrm{obj}(\mathcal{C})$, pre-composition with f defines a bijection

$$f_* : \mathcal{C}(y, c) \to \mathcal{C}(x, c).$$

Remark 1.2.4 (Note on representability)**.**

Definition 1.2.7 (Monomorphism, epimorphism)**.**

$$z \xrightarrow{\text{epi}} x \underset{g}{\overset{f}{\rightrightarrows}} y \xrightarrow{\text{mono}} w \ .$$

(to remember: epi $\to$ mono)

Example 1.2.8 (Mono in $\mathcal{S}et$ are precisely the injections)**.**

Example 1.2.9 (Section (right inverse), retraction (left inverse); split mono / epi)**.**

Example 1.2.10 (Iso $\Rightarrow$ mono and epi; counterexample $\mathbb{Z} \hookrightarrow \mathbb{Q}$ in $\mathcal{R}ing$ for the converse)**.**

Lemma 1.2.11 (Compositions of mono / epi)**.**

1. Composable f and g are mono $\Rightarrow$ gf is mono. Dually, f and g are epi $\Rightarrow$ gf is epi.
2. gf is mono $\Rightarrow$ f is mono. Dually, gf is epi $\Rightarrow$ g is epi.

1.3 Functoriality

Definition 1.3.1 (Functor)**.** Notation: $F : \mathcal{C} \to \mathcal{D}$.

Example 1.3.2 (Functors)**.**

- Power set functor $P : \mathcal{S}et \to \mathcal{S}et$;
- Forgetful functors $U : \mathcal{C} \to \mathcal{S}et$, itermediate forgetful functors (e.g. $\mathcal{M}od_R \to \mathcal{A}b$, $\mathcal{G}r \to \mathcal{S}et_*$);
- Homotopy class functors $\mathcal{T}op_{(*)} \to \mathcal{H}tpy_{(*)}$;
- Fundamental group $\pi_1 : \mathcal{T}op_* \to \mathcal{G}rp$ and its factor $\pi_1 : \mathcal{H}tpy_* \to \mathcal{G}rp$ through $\mathcal{T}op_* \to \mathcal{H}tpy_*$, fundamental groupoid $\Pi_1 : \mathcal{T}op \to \mathcal{G}roupoid$;
- n-cycles Z_n, n-boundary B_n and n-th homology H_n typed $\mathcal{C}h_R \to \mathcal{M}od_R$, all degrees simultaneously give $Z_*, B_*, H_* : \mathcal{C}h_R \to \mathcal{G}r\mathcal{M}od_R$ (codomain is the category of graded R-modules);
- Free functors $F : \mathcal{S}et \to \mathcal{C}$ (left adjoints to forgetful functors);
- Derivative $D : \mathcal{E}uclid_* \to \mathcal{M}at_{\mathbb{R}}$;
- $M^- : \mathcal{F}in_* \to \mathcal{S}et$ for a commutative monoid M.

Theorem 1.3.3 (Brouwer Fixed Point Theorem)**.** Any continuous endomorphism of a 2-dimensional disk D^2 has a fixed point.

Example 1.3.4 (Example for topological data analysis)**.**

Definition 1.3.5 (Contravariant functor)**.** $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Example 1.3.7 (Contravariant functors)**.**

- Contravariant power set $P : \mathcal{S}et^{\mathrm{op}} \to \mathcal{S}et$;
- Dual vector space $* : \mathcal{V}ect_{\mathbf{k}}^{\mathrm{op}} \to \mathcal{V}ect_{\mathbf{k}}$;
- Continuity $O : \mathcal{T}op^{\mathrm{op}} \to \mathcal{P}oset$;
- Zarisky topology $\mathcal{CR}ing \to \mathcal{T}op$;
- Presheafs $\mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$ and simplex category.

Lemma 1.3.8 (Functors preserve isomorphisms)**.**

Example 1.3.9 (Functors $\mathcal{B}G \to \mathcal{C}$ ($\mathcal{B}G^{\mathrm{op}} \to \mathcal{C}$) are left (right) group actions)**.**

Corollary 1.3.10 (Functorial action of a group)**.**

Definition 1.3.11 (Partial hom functors)**.** $\mathcal{C}(c, -) : \mathcal{C} \to \mathcal{S}et$, $\mathcal{C}(-, c) : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$ for a locally small $\mathcal{C}$.

Definition 1.3.12 (Product of categories)**.** Notation: $\mathcal{C} \times \mathcal{D}$.

Definition 1.3.13 (Hom functor)**.** $\mathcal{C}(-, -) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{S}et$.

Definition (Categories of categories)**.**

- $\mathcal{C}at$ – category of small categories;
- $\mathcal{CAT}$ – category of locally small categories.

Example 1.3.14 (Isomorphisms of categories)**.**

- $(-)^{\mathrm{op}} : \mathcal{CAT} \to \mathcal{CAT}$;
- $(-)^{-1} : \mathcal{B}G \to \mathcal{B}G^{\mathrm{op}}$;
- Groupoid is isomorphic to its opposite;
- $\mathcal{M}od_R \simeq {}_{R^{\mathrm{op}}}\mathcal{M}od$;
- $\mathcal{M}at_R \simeq \mathcal{M}at_R^{\mathrm{op}}$.

Example 1.3.15 (Galois extension)**.**

1.4 Naturality

Definition 1.4.1 (Natural transformation, natural isomorphism)**.** Notation: $\alpha : F \to G$.

Example 1.4.3 (Natural transformations)**.**

- $\mathrm{ev} : \mathrm{id}_{\mathcal{V}ect} \to **$, iso on $\mathcal{F}d\mathcal{V}ect$;
- G-equivariant map $\mathcal{B}G \to \mathcal{C}$;

– Open and closed subset functors $O, C : \mathcal{T}op^{\mathrm{op}} \to \mathcal{S}et$ are naturally isomorphic;

Proposition 1.4.4 (Example of an iso that is not natural (torsion subgroup))**.**

Example 1.4.5 (Riesz representation theorem)**.**

Example 1.4.6 (Natural transformation between hom functors)**.**

Example 1.4.8 (Categorification of natural numbers)**.**

1.5 Equivalence of categories

Lemma 1.5.1 (Internal representation of natural transformations)**.** Fixing a parallel pair of functors $F, G : \mathcal{C} \to \mathcal{D}$, natural transformations $\alpha : F \to G$ correspond bijectively to functors $H : \mathcal{C} \times 2 \to \mathcal{D}$ such that

$$\begin{tikzcd} \mathcal{C} \arrow[r, "i_0"] \arrow[dr, "F"'] & \mathcal{C} \times 2 \arrow[d, "H"] & \mathcal{C} \arrow[l, "i_1"'] \arrow[dl, "G"] \\ & \mathcal{D} & \end{tikzcd} .$$

Definition 1.5.4 (Equivalence of categories)**.** Notation: $\mathcal{C} \sim \mathcal{D}$.

Lemma 1.5.5 (Equivalence of categories defines an equivalence relation)**.**

Example 1.5.6 (Equivalence between the category of pointed sets and the category of sets with partial functions)**.**

Definition 1.5.7 (Full, faithful, essentially surjective on objects functors)**.**

Remark 1.5.8 ((Full) embedding, full subcategory)**.**

Theorem 1.5.9 (Characterizing equivalence of categories)**.** A functor defininig an equivalence of categories is full, faithful and essentially surjective on objects. Assumming the axiom of choice, any functor with these properties defines an equivalence of categories.

Corollary 1.5.11 (An equivalence between abstract and concrete linear algebra)**.**

Proposition 1.5.12. Any connected groupoid is equivalent to the automorphism group of any of its objects.

Corollary 1.5.13. In a path-connected space X, any choice of basepoint $x \in X$ yields an isomorphic fundamental group $\pi_1(X, x)$.

Remark 1.5.14. Frequently, one functor of an equivalence of categories can be defined canonically, while the inverse requires the axiom of choice.

Definition 1.5.15 (Skeleton and skeletal category)**.** Notation: $\mathrm{sk}\mathcal{C}$.

Remark 1.5.16 (Construction of skeleton)**.**

Example 1.5.17 (Skeletons)**.**

Example 1.5.18 (A categorification of the orbit-stabilizer theorem)**.**

1.6 The art of the diagram chase

Definition 1.6.2 (Monoid object)**.**

- Monoid in $\mathcal{S}et$;
- Topological monoid;
- Unital ring;
- $\mathbf{k}$-algebra.

Lemma 1.6.4 (Functors preserve commutative diagrams)**.**

Lemma 1.6.11 (Rewriting in compositions of morphisms)**.**

$$f_j \dots f_k = g_m \dots g_1 \Rightarrow f_1 \dots f_j \dots f_k \dots f_n = f_1 \dots g_m \dots g_1 \dots f_n.$$

Lemma 1.6.12 (Reversing iso on the diagram)**.**

Lemma 1.6.13 (Reversing multimple isos on the diagram)**.**

Definition 1.6.14 (Initial and terminal objects)**.**

Example 1.6.15 (Examples of initial and terminal objects)**.**

- Empty set / one-element set in $\mathcal{S}et$;
- Empty and singleton space in $\mathcal{T}op$;
- Every one-element set in $\mathcal{S}et_*$;
- Zero module is both initial and terminal in $\mathcal{M}od_R$, the trivial group is both initial and terminal in $\mathcal{G}r$;
- $\mathbb{Z}$ is the initial object in $\mathcal{R}ing$;
- $\mathcal{F}ield$ has neigher initial nor terminal objects;
- The empty category is initial and the category 1 is terminal in $\mathcal{C}at$;
- A global minimal and a global maximal elements (if they exist) are the initial and terminal ojbects in a preorder.

Lemma 1.6.16 (Compositions from initial / to terminal objects)**.**

Definition 1.6.17 (Concrete category)**.** A concrete category is a category $\mathcal{C}$ equipped with a faithful functor $U : \mathcal{C} \to \mathcal{S}et$.

Lemma 1.6.18 (Faithful preimage of a commutative diagram)**.** $U : \mathcal{C} \to \mathcal{D}$ is faithful $\Rightarrow$ preimage of commutative diagram in $\mathcal{D}$ is commutative in $\mathcal{C}$.

Remark 1.6.20 (On cancellability)**.** Adjacent squares commute $\Rightarrow$ external square commutes. Converse need not hold.

Lemma 1.6.21. Consider morphisms with the indicated sources and targets

$$\begin{tikzcd} a \arrow[r, "f"] \arrow[d, "g"'] & b \arrow[r, "j"] \arrow[d, "h"'] & c \arrow[d, "t"'] \\ a' \arrow[r, "k"'] & b' \arrow[r, "m"'] & c' \end{tikzcd}$$

and suppose that the outer rectangle commutes. This data defines a commutative rectangle if either:

(i) the right-hand square commutes and m is a mono;

(ii) the left-hand square commutes and f is an epi.

1.7 The 2-category of categories

Lemma 1.7.1 (Vectical composition)**.** $\alpha : F \to G$, $\beta : G \to H \Rightarrow$

$$\beta \circ \alpha : F \to H, \quad (b \circ \alpha)_c := \beta_c \circ \alpha_c$$

Corollary 1.7.2 (Functor category $\mathcal{D}^{\mathcal{C}}$)**.**

Remark 1.7.3 (Sizes of functor categories)**.**

Lemma 1.7.4 (Horizontal composition)**.** $\alpha : F \to G$, $\beta : H \to K \Rightarrow$

$$\beta \bullet \alpha : HF \to KG, \quad (\beta \bullet \alpha)_c = K\alpha_c \circ \beta_{Fc} = \beta_{Gc} \circ H\alpha_c$$

Remark 1.7.6 (Whiskering)**.** $L\beta F : LHF \to LKF$:

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{E} \arrow[r, "L"] & \mathcal{F} \end{tikzcd}$$

Lemma 1.7.7 (Middle four interchange)**.**

$$(\delta \circ \gamma) \bullet (\beta \circ \alpha) = (\delta \bullet \beta) \circ (\gamma \bullet \alpha)$$

Definition 1.7.8 (2-category)**.**

- Objects (0-morphism) $c, d : \mathcal{C}$;
- 1-morphsism between objects $f : c \to d$;
- 2-morphisms between parallel pairs of 1-morphisms $\alpha : f \to g$;
- + Axioms (in particular, middle four interchange).

2 Universal Properties, Representability and the Yoneda Lemma

Illustration: coloring of the graph.

$$n\text{-Color} : \text{Graph}^{\text{op}} \to \mathcal{S}et$$

$$\begin{tikzcd} G \arrow[r, "\text{coloring}"] & \text{discr}(n) \\ G' \arrow[u, "f"] \arrow[ur, "\text{coloring}'"'] & \end{tikzcd}$$

K_n represents the n-Color:

$$\text{Hom}(-, K_n) \simeq n\text{-Color}$$

+ in 2.3 we define the notion of the universal element that witnesses a universal property of some object in a locally small category. The universal element witnessing the universal property of the complete graph is an n-coloring of K_n, an element of the set n-Color(K_n). In 2.4, we use the Yoneda lemma to show that the pair comprised of an object characterized by a universal property and its universal element defines a universal object in a category of elements of the functor that it represents.

2.1 Representable functors

Example 2.1.1 (Discrete dynamic system)**.** $(X, f : X \to X, x_0 \in X)$. $f(x_n) = x_{n+1}$. Natural numbers is the universal discrete dynamic system. Consider the category with objects $(1 \xrightarrow{q} X \xrightarrow{f} X)$; the $(1 \xrightarrow{0} \mathbb{N} \xrightarrow{s} \mathbb{N})$ is its initial object:

$$\begin{tikzcd} 1 \arrow[r, "0"] \arrow[dr, "q"'] & \mathbb{N} \arrow[r, "s"] \arrow[d, dashed, "r"] & \mathbb{N} \arrow[d, dashed, "r"] \\ & X \arrow[r, "f"] & X \end{tikzcd} \qquad \begin{tikzcd} * \arrow[r, mapsto] \arrow[dr, mapsto] & 0 \arrow[r, mapsto] \arrow[d, mapsto] & s(0) \arrow[d, mapsto] \\ & q = r(0) \arrow[r, mapsto] & f(q) = r(s(0)) \end{tikzcd}$$

Definition 2.1.3 (A representable characterization of initial or terminal objects)**.**

- c in $\mathcal{C}$ is initial iff $\mathcal{C}(c, -) : \mathcal{C} \to \mathcal{S}et$ is isomorphic to a constant functor $\lambda - .\{*\} : \mathcal{C} \to \mathcal{S}et$
- $c \in \mathcal{C}$ is terminal iff the functor $\mathcal{C}(-, c) : \mathcal{C}^{\text{op}} \to \mathcal{S}et$ is iso to a functor $\lambda - .\{*\} : \mathcal{C}^{\text{op}} \to \mathcal{S}et$

Definition 2.1.4 (Representable functor and representation)**.**

- $F : \mathcal{C} \to \mathcal{S}et$ is representable if isomorphic to $\mathcal{C}(c, -)$ for some $c : \text{obj}(\mathcal{C})$
- Representation – a choice of object with a specified isomorphism.

Example 2.1.5 (Covariant representable functors)**.** 1. $\text{id}_{\mathcal{S}et}$ by a singleton;

2. $U : \mathcal{G}rp \to \mathcal{S}et$ by group $\mathbb{Z}$;
3. $U : \mathcal{M}od_R \to \mathcal{S}et$ is represented by the R-module R (R - unital);
4. $U : \mathcal{R}ing \to \mathcal{S}et$ is represented by the unital ring $\mathbb{Z}[x]$;
5. (group presentation);
6. $(-)^X : \mathcal{R}ing \to \mathcal{S}et$ is represented by Laurent polynomials $\mathbb{Z}[x, x^{-1}]$;
7. The forgetful functor $U : \mathcal{T}op \to \mathcal{S}]\sqcup$ by the singleton space;
8. ($\mathcal{C}at$ stuff);
9. $U : \mathcal{S}et_* \to \mathcal{S}et$;
10. $\mathcal{P}ath : \mathcal{T}op \to \mathcal{S}et$.

Example 2.1.6 (Contravariant representable functors)**.**

1. Power set $P : \mathcal{S}et^{\mathrm{op}} \to \mathcal{S}et$, represented by $\{\top, \bot\}$;
2. $O : \mathcal{T}op^{\mathrm{op}} \to \mathcal{S}et$ that sends a space to its set of open subsets is represented by the Sierpinski space S – a space with two points, one closed and one open. S also represents $C : \mathcal{T}op^{\mathrm{op}} \to \mathcal{S}et$ (closed subsets);
3. $\mathrm{Hom}(- \times A, B) : \mathcal{S}et^{\mathrm{op}} \to \mathcal{S}et$ is represented by the set B^A of functions from A to B;
4. $U(-)^* : \mathrm{Vect}_{\mathbf{k}}^{\mathrm{op}} \to \mathcal{S}et$ is represented by $\mathbf{k}$;
5. (topology / homological algebra)

2.2 The Yoneda lemma

Q: Given a functor $F : \mathcal{C} \to \mathcal{S}et$, what data is needed to define a morphism $\mathcal{C}(c, -) \to F$?

Example 2.2.1 (Ordinal category)**.**

Example 2.2.2 ($\mathcal{B}G$)**.**

Proposition 2.2.3. G-equivariant maps $G \to X$ correspond bijectively to elements of X.

Theorem 2.2.4 (Yoneda lemma)**.** For any $F : \mathcal{C} \to \mathcal{S}et$, whose domain $\mathcal{C}$ is locally small and any object $c : \mathrm{obj}(\mathcal{C})$, there is a bijection

$$\mathrm{Hom}(\mathcal{C}(c, -), F) \cong Fc$$

that associates a natural transformation $\alpha : \mathcal{C}(c, -) \to F$ to the element $\alpha_c(\mathrm{id}_c)$. Morover, this correspondence is natural in both c and F.

Remark 2.2.7 (Size issues vs. concise formulation)**.**

$$\mathrm{Hom}(y(-), -) := \mathcal{C} \times \mathcal{S}et^{\mathcal{C}} \to (\mathcal{S}et^{\mathcal{C}})^{\mathrm{op}} \times \mathcal{S}et^{\mathcal{C}} \xrightarrow{\mathrm{Hom}} \mathcal{SET}$$

Though everything is fine with the composite. Some unclear notation here in general; can be written this way (here A, B denotes a hom-set):

$$\begin{tikzcd} \mathcal{C} \times \mathcal{S}et^{\mathcal{C}} \arrow[r, bend left, "\lambda AF.\mathcal{C}(A,_),F"] \arrow[r, bend right, "\lambda AF.FA"'] \arrow[r, phantom, "\Downarrow\simeq"] & \mathcal{S}et \end{tikzcd} \quad \Bigg| \quad \begin{tikzcd} \mathcal{C}^{\mathrm{op}} \times \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}} \arrow[r, bend left, "\lambda AF.\mathcal{C}(_,A),F"] \arrow[r, bend right, "\lambda AF.FA"'] \arrow[r, phantom, "\Downarrow\simeq"] & \mathcal{S}et \end{tikzcd}$$

Essentially this says that natural transformations $\mathcal{C}(A, -) \to F$ correspond to evaluations $F(A)$. More precise typing:

$$\lambda fF.[\lambda g.\mathcal{C}(f, g), F] \to \lambda fF.F(f).$$

Corollary 2.2.8 (Yoneda embedding)**.**

$$Y = \lambda f.\mathcal{C}(-, f) : \mathcal{C} \to \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}}, \quad Y' = \lambda f.\mathcal{C}(f, -) : \mathcal{C} \to \mathcal{S}et^{\mathcal{C}},$$

$$\begin{tikzcd} c \arrow[d, "f"'] & \mathrm{Hom}(-, c) : \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}} \arrow[d, "\mathrm{Hom}(-,f)"'] & \mathrm{Hom}(c, -) : \mathcal{S}et^{\mathcal{C}} \\ d & \mathrm{Hom}(-, d) : \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}} & \mathrm{Hom}(d, -) : \mathcal{S}et^{\mathcal{C}} \arrow[u, "\mathrm{Hom}(f,-)"] \end{tikzcd}$$

define full and faithful embeddings. Here objects c, d in Hom functor expressions are indentified with id_c, id_d. Note, that this is essentially obtained by 2-categorical currying adjunction.

Explicitly:

$$c \xrightarrow{\quad f \quad} d$$

$$\begin{tikzcd} x \arrow[d, "g"'] & \mathrm{Hom}(x, c) \arrow[r, "f\circ-"] & \mathrm{Hom}(x, d) & \mathrm{Hom}(c, x) \arrow[d, "g\circ-"'] & \mathrm{Hom}(d, x) \arrow[l, "-\circ f"'] \arrow[d, "g\circ-"] \\ y & \mathrm{Hom}(y, c) \arrow[u, "-\circ g"] \arrow[r, "f\circ-"'] & \mathrm{Hom}(y, d) \arrow[u, "-\circ g"'] & \mathrm{Hom}(c, y) & \mathrm{Hom}(d, y) \arrow[l, "-\circ f"] \end{tikzcd} .$$

The fully faithfullness:

$$\mathcal{C}(c, d) \xrightarrow{\cong} \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}}(\mathcal{C}(-, c), \mathcal{D}(-, d)), \quad \mathcal{C}(c, d) \xrightarrow{\cong} \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}}(\mathcal{C}(d, -), \mathcal{D}(c, -)).$$

Corollary 2.2.9 (Matrices)**.**

Corollary 2.2.10 (Permutation group)**.** Any group is isomorphic to a subgroup of a permutation group.

(note: Emiliy Riehl's proof contains a lot of handwaving, and the typing is not clear; mine requires polish too)

Proof. Consider $\mathcal{B}G$. The image of its single object under the covariant Yoneda embedding $Y : \mathcal{B}G \to \mathcal{S}et^{\mathcal{B}G^{\mathrm{op}}}$ is the right G-set:

$$\begin{tikzcd} G \arrow[d, "h"'] & \mathcal{B}G(-,G) \\ G & \mathcal{B}G(-,G) \arrow[u, "h\circ -"'] \end{tikzcd} \quad ; \quad \begin{tikzcd} G \arrow[d, "g"'] & \mathcal{B}G(G,G) \arrow[r, "h\circ -"] & \mathcal{B}G(G,G) \\ G & \mathcal{B}G(G,G) \arrow[u, "-\circ g"] \arrow[r, "h\circ -"'] & \mathcal{B}G(G,G) \arrow[u, "-\circ g"'] \end{tikzcd} \quad \begin{tikzcd} G \arrow[r, "h"] & G \end{tikzcd} .$$

The functoriality of Y implies that the images of the arrows of $\mathcal{B}G$ are automorphisms. The fully faithfullness of Y (Cor. 2.2.8)

$$\mathcal{B}G(G,G) \xrightarrow{\cong} \mathcal{S}et^{\mathcal{B}G^{\mathrm{op}}}(\mathcal{B}G(-,G), \mathcal{B}G(-,G)).$$

implies that all of the endomorphisms of $\mathcal{B}G(-,G)$ (equivariant maps) are automorphisms and correspond to the arrows of $\mathcal{B}G$ (elements of group G). Concretely, they are given by left multiplication with elements of G.

So, Yoneda embedding defines a bijection between G and G-set G. The bijection extends to an isomorphism on $\mathcal{G}r$ due to the functoriality of Y, and $\mathrm{Aut}_{G\text{-}\mathcal{S}et}(G)$ is a subgroup of $\mathrm{Sym}(G)$:

$$G \simeq \mathrm{Aut}_{G\text{-}\mathcal{S}et}(G) \hookrightarrow \mathrm{Sym}(G).$$

□

2.3 Universal properties and universal elements

In a locally small $\mathcal{C}$, $x \simeq y$ are represetably isomorphic (by functoriality):

$$\mathcal{C}(-,x) \cong \mathcal{C}(-,y), \quad \mathcal{C}(x,-) \cong \mathcal{C}(y,-).$$

Proposition 2.3.1. Isomorphisms of representable objects $\Leftrightarrow$ isomorphism of functors represented by them.

Corollary 2.3.2. The full subcategory of $\mathcal{C}$ spanned by its terminal objects is either empty or is a contractible groupoid. In particular, any two terminal objects in $\mathcal{C}$ are uniquely isomorphic.

Definition 2.3.3 (Universal property, universal element)**.** A universal property of an object $c \in \mathcal{C}$ is expressed by a representable functor F together with a universal element $x \in Fc$ that defines a natural iso $\mathcal{C}(c,-) \simeq F$ (or $\mathcal{C}(-,c) \simeq F$).

Example 2.3.4. For $U : \mathcal{R}ing \to \mathcal{S}et$, $U \simeq \mathcal{R}ing(\mathbb{Z}[x],-)$:

$$\mathcal{R}ing(\mathbb{Z}[x],R) \cong UR,$$

$$\begin{tikzcd}
\mathbb{Z}[x] \arrow[d, "f"'] & \mathcal{R}ing(\mathbb{Z}[x],\mathbb{Z}[x]) \arrow[r, "\alpha_{\mathbb{Z}[x]}"] \arrow[d, "f\circ -"'] & U\mathbb{Z}[x] \arrow[d, "Uf"] & \mathrm{id}_{\mathbb{Z}[x]} \arrow[r, mapsto] \arrow[d, mapsto] & \alpha_{\mathbb{Z}[x]}(\mathrm{id}_{\mathbb{Z}[x]}) \arrow[d, mapsto] \\
R & \mathcal{R}ing(\mathbb{Z}[x],R) \arrow[r, "\alpha_R"'] & UR & f \arrow[r, mapsto] & \alpha_R(f) = Uf(\alpha_{\mathbb{Z}[x]}(\mathrm{id}_{\mathbb{Z}[x]}))
\end{tikzcd}.$$

The natural iso is determined by the image of $\mathrm{id}_{\mathbb{Z}[x]}$.

Example 2.3.6. $E : \mathcal{B}G \to \mathcal{S}et$ is representable iff there is an isomorphism $G \simeq E$ of left G-sets. This implies that the action of G on E is free and transitive. A representable G-set is called a G-torsor.

Example 2.3.7. For $\mathrm{Bilin}(V, W; -) : \mathcal{V}ect_{\mathbf{k}} \to \mathcal{S}et$,

$$\mathcal{V}ect_{\mathbf{k}}(V \otimes W, U) \cong \mathrm{Bilin}(V, W; U).$$

Proposition 2.3.9 (Symmetry of $\otimes$)**.**

Remark 2.3.11 (Explicit symmetry iso)**.**

Example 2.3.12 (Group ring)**.** $\mathbb{Z}[G]$,

$$\mathcal{R}ing \xrightarrow{\mathrm{Group}(G,(-)^\times)} \mathcal{S}et.$$

2.4 Category of elements

A universal property for $c : \mathcal{C}$ is expressed by $(c, \alpha : F \xrightarrow{\simeq} \mathcal{C}(c, -))$ (covariant). Natural characterization of maps into / out of c. The main result of this section proves that any universal property can be understood as defining an initial or terminal object.

Remark. The flow is weird, imo. The structural definition of universal objects is more fundamental, while Riehl tells the story in reverse order.

Q:

1. How to determine whether a given set-valued functor F is representable?
2. (!) Is there a way to identify when an element $x \in Fc$ determines a natural iso, rather than a natural transformation?

Definition 2.4.1 (Category of elements)**.** The category of elements $\int F$ of a covariant functor $F : \mathcal{C} \to \mathcal{S}et$ is defined as:

- Objects: $(c : \mathcal{C}, x \in Fc)$;
- Arrows: $(c, x) \to (c', x')$ is a morphism $f : c \to c'$ s.t. $Ff(x) = x'$.

$$\begin{tikzcd}
c \arrow[d, "f"'] & (c,x) \arrow[d, "f"'] & Fc \arrow[d, "Ff"'] & x \arrow[d, mapsto, "Ff"'] \\
c' & (c',x') & Fc' & f(x)
\end{tikzcd}.$$

Forgetful functor $\Pi : \int F \to \mathcal{C}$:

$$(c, x) \xrightarrow{f} (c', x') \quad \mapsto \quad c \xrightarrow{f} c'.$$

Definition 2.4.2 (The category of elements of a contravarian functor)**.** The category of elements $\int F$ of a contravariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{S}et$ is defined as:

- Objects: $(c : \mathcal{C}, x \in Fc)$;
- Arrows: $(c, x) \to (c', x')$ is a morphism $f : c \to c'$ s.t. $Ff(x') = x$.

Forgetful functor works similarly.

Example 2.4.3 ($\int n$-Color)**.** $\int n$-Color is the category of n-colored graphs and color preserving graph homomorphisms.

Example 2.4.4 ($U : \mathcal{C} \to \mathcal{S}et$)**.** $\int U$ is the category of based objects in $\mathcal{C}$. Note that only functions in $\mathcal{C}$ are allowed.

Example 2.4.5 (Dependent sum and dependent product)**.** When $\mathcal{C}$ is a discrete category, a functor $F : \mathcal{C} \to \mathcal{S}et$ encodes an indexed family of sets $(F_c)_{c \in \mathcal{C}}$. $\int F$ is a discrete category whose set of objects in $\coprod_{c:\mathcal{C}} F_c$ – dependent sum of the indexed family of sets $(F_c)_{c:\mathcal{C}}$. The canonical projection $\Pi : \coprod_{c:\mathcal{C}} F_c \to \mathcal{C}$ defined so that the fiber over $c : \mathcal{C}$ is the set F_c. The related dependent product $\prod_{c:\mathcal{C}} F_c$ is the set of sections of the functor $\Pi : \coprod_{c:\mathcal{C}} F_c \to \mathcal{C}$.

Example 2.4.6 (Slice and coslise for $\int \mathcal{C}(c, -)$ and $\int \mathcal{C}(-, c)$)**.** Morphisms of $\int \mathcal{C}(c, -)$ are

$$f : (d, h \in \mathcal{C}(c, d)) \to (d', h' \in \mathcal{C}(c, d')),$$

such that $\mathcal{C}(c, -)(f)(h) = f \circ h = h'$ (note: polymorphic typing of f):

$$\begin{tikzcd} & c \arrow[dl, "h'"'] \arrow[dr, "h"] & \\ d' & & d \arrow[ll, "f"] \end{tikzcd} .$$

So, $\int \mathcal{C}(-, c)$ is the slice category $c/\mathcal{C}$ over the object $c : \mathcal{C}$. Dually, $\int \mathcal{C}(c, -)$ is the coslise category $\mathcal{C}/c$.

Lemma 2.4.7. For $F : \mathcal{C}^o p \to \mathcal{S}et$, the category of elements is isomorphic to the comma category:

$$\int F \simeq (y \downarrow F)$$

defined relative to the Yoneda embedding $y : \mathcal{C} \to \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}}$ and the object $F : 1 \to \mathcal{S}et^{\mathcal{C}^{\mathrm{op}}}$.

Proposition 2.4.8. A covariant set-valued functor is representable iff its category of elements has an initial object. Dually, a contravariant set-valued functor is representable iff its category of elements has a terminal object.

(the subsection is unfinished)

3 Limits and Colimits